SpeedUp v2 User Check

Visual decisions.
Measured motion fragments.
One persistent task owner.

A model-led robot executive that delegates repeated control while retaining the ability to recover and revise its plan.





Recorded initial scenes
t3: bowl into bottom drawer, then close.
t6: mug onto plate; pudding to its right.

English edition · October 21, 2026
gpt-6-astra / high · LIBERO-10 t3 and t6

RESULTS AT A GLANCE

What the current evidence shows

One matched initialization per task; the prior reference is v6 User Check, not a new Astranomy benchmark.

2.49×

t3 observed speed at raw goal completion

99%+

v7 actions executed by generic fragments

Incomplete

t6 stopped by user; no completion ratio

t3: faster, with quality caveats

8:29 versus 21:10; 22 versus 87 model decisions.
Raw LIBERO success in both runs.
Strict physical completion fails in both.

t6: execution works; goal still missed

Mug placed; pudding grasped and repositioned.
Stopped at 12:35 and 28 decisions.
Final pudding center lies outside target region.

These runs demonstrate controller use and a t3 timing improvement, not broad task generalization.

[R1–R4] Saved summaries, agent_stats and physical_truth. Times exclude video rendering and initial plan preparation.

DESIGN PRINCIPLE

The model remains responsible for the task

Use fast execution where the target is clear; return control when the execution evidence changes.

Model responsibility

Interpret the task and scene.
Choose grasp, route and object goal.
Inspect outcomes.
Change method when needed.

Executor responsibility

Track numerical targets from measured state.
Run explicit gripper stages.
Detect local completion or lack of progress.
Return measured evidence.

Independent evaluation

Record benchmark success.
Audit release, support and stability separately.
Record contacts.
Do not feed hidden truth to the policy.

Connection to Astranomy: retain model-directed numerical control and visual correction; v7 adds longer measured execution sequences between model decisions.

[M1–M4] agent.py, agent_client.py, fragments.py, sim.py. Existing Astranomy and original v2 implementations are unchanged.

ARCHITECTURE

The full control loop

Visual reasoning between fragments; proprioceptive feedback control within fragments.

Initial review

Task + draft JSON
Object / goal selectors
One-time approval

→

Persistent executive

Current RGB + robot state
Reviewed goals / constraints
High-effort model decision

→

Execution branch

execute: fragments
motion: short target
macro: optional native skill

→

Measured handoff

Pose / gap / action result
Completed / pending stages
Fresh visual observation

Arrival, checkpoint, stall or exhausted budget returns control to the same executive.

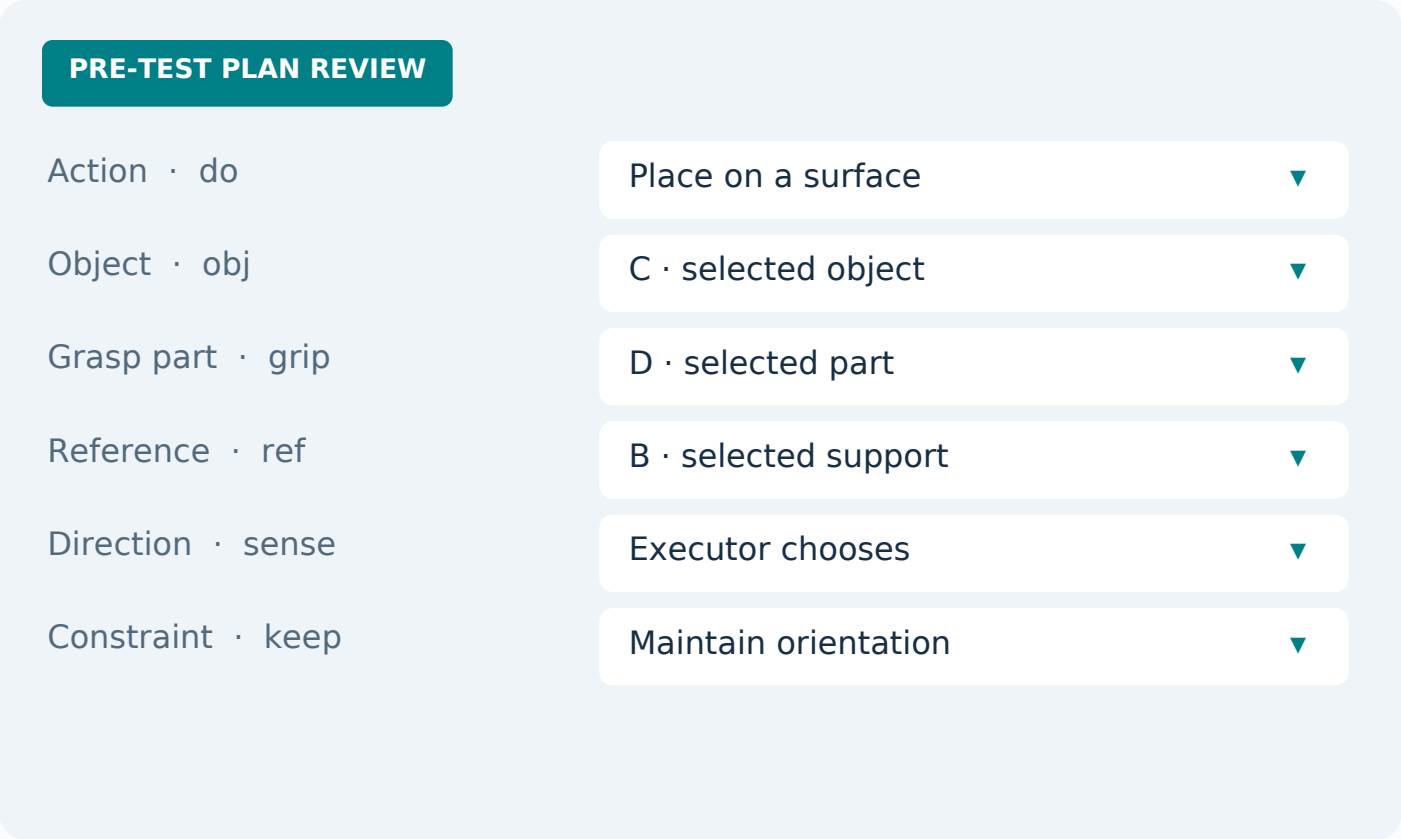
Independent benchmark and physical audit observe execution outside this control loop.

[M1,M2,M3] A finish response is a model claim plus fresh observation; it cannot set the benchmark success flag.

BEFORE EACH TEST

The English review interface

Selection-based editing generates the JSON; the user does not need to type a program.



English throughout

Headings, choices, buttons, status messages, errors and accessibility labels.

Review scope

Objects, task steps, preserved relations, spatial regions and placement goals.

Recent v7 tests used explicitly authorized automatic approval; no manual review occurred.

[UI] agent_speedup_v2_user_check.html. Left: schematic of the actual selection fields, not a browser screenshot.

ACTION INTERFACE

Three execution paths, different responsibilities

The model selects one action type per decision. Optional payloads not used by that type remain null.

Path	What it executes	What this does not imply
execute	1–6 generic segments; measured completion; up to 240 control steps per sequence.	No object recognition, grasp certificate or collision-free path guarantee.
motion	A numerical pose/gripper target for 1–5 control steps.	Target may not be reached within that short grant.
macro	Optional legacy hold, turn, place_on or release, with its own prerequisites.	A semantic plan intent is not proof that a native macro implements it.
observe / finish	Fresh observation, or a completion claim followed by observation.	Neither action sets official success.

v7’s measured acceleration in these tests came from execute fragments; native macro actions were zero.

[M1,M2,M3,M5] Legacy macro grants are bounded and may require calibration / grounding; they are optional.

REAL EXECUTION EXAMPLE

One decision can close, test-lift and observe

t6_009, decision 3: the model supplies the sequence; the same generic executor handles both stages.


```
"execution": {
  "segments": [
    {"skill": "set_gripper",
     "xyz_m": null, "quaternion_wxyz": null,
     "gripper": "close", "profile": "contact",
     "max_steps": 40, "observation_after": false},
    {"skill": "move_relative",
     "xyz_m": [0, 0, 0.03],
     "quaternion_wxyz": null, "gripper": "keep",
     "profile": "contact", "max_steps": 55,
     "observation_after": true}
  ]
}
```


Explicit stages

Close until local measured conditions pass, then lift from the actual segment-entry pose.

Visual confirmation belongs to the model

After the lift, inspect whether the mug moved with the gripper. Stable gap alone is insufficient.

Shown: execution payload excerpt.

[R4 request 3; M3] The 3 cm lift is a model-selected value from this run, not a task-specific constant in the executor.

LOCAL CONTROL

Measured completion and explicit return points

The controller converts pose error into bounded OSC commands at 20 Hz.

Design bound	Transit profile	Contact profile
Target translation from entry	Up to 30 cm	Up to 8 cm
Per-action translation cap	2 cm	5 mm
Position arrival tolerance	4 mm	2 mm
Target rotation / action cap	1.2 rad / 0.1 rad	0.5 rad / 0.1 rad

Advance on measured completion

Relative goals use actual entry pose; gripper open/close is a separate stage.

Return on uncertainty or boundary

Stall, gap change, envelope rejection, budget or observation checkpoint returns control.

[M3] Max 120 steps/segment, 240/sequence. Contact is a slower profile—not contact sensing or a safety certificate.

GENERALITY AND INFORMATION BOUNDARY

What is reusable—and what still requires judgment

Task-independent code does not establish that all tasks or spatial meanings are supported correctly.

Reusable implementation

Five fragments: move_to_pose, move_relative, approach, set_gripper, wait.

Measured state and explicit handoffs.
No task IDs, object names or predefined task coordinates in the fragment executor.

Model-selected route, grasp and checkpoints.

Limits of the current interface

Input: RGB, robot pose/gap/joints, camera and tool geometry, action feedback.

No hidden object poses, depth, contact identities or goal predicates.

Fragment constraints rely on model interpretation; there is no whole-arm collision planner.

Local feedback is proprioceptive. Fresh visual reasoning happens at model decision boundaries.

[M1–M4] Six recent image pairs plus continuous textual execution history. Geometry calibration does not provide object depth.

EXPERIMENT SETUP

Real-task evaluation protocol

All tests use existing LIBERO scenes; no synthetic trajectory or replay is substituted for a task result.

Setting	Recorded value
Model / effort	gpt-6-astra / high; subscription route; zero API-key calls
Matched conditions	LIBERO-10 t3 or t6; episode 0; seed 7; matching initial observation hashes within each task
Reference	Prior v6 User Check: t3_002 and t6_007; not a fresh pure-Astranomy comparison
Current trials	v7 t3_003; t6_009 after user-requested restart of the stalled t6_008 preparation
Approval	Prior runs manually reviewed; v7 initial plans automatically approved with user authorization
Outcome / timing	Raw LIBERO goal; separate strict physical audit; contacts recorded. Control time excludes video and preparation.

[R1–R5] Frozen run/review records. Local implementation validation was limited; no repeated smoke rollout was performed.

RECORDED SCENE / T3_003

t3: bowl into the bottom drawer, then close

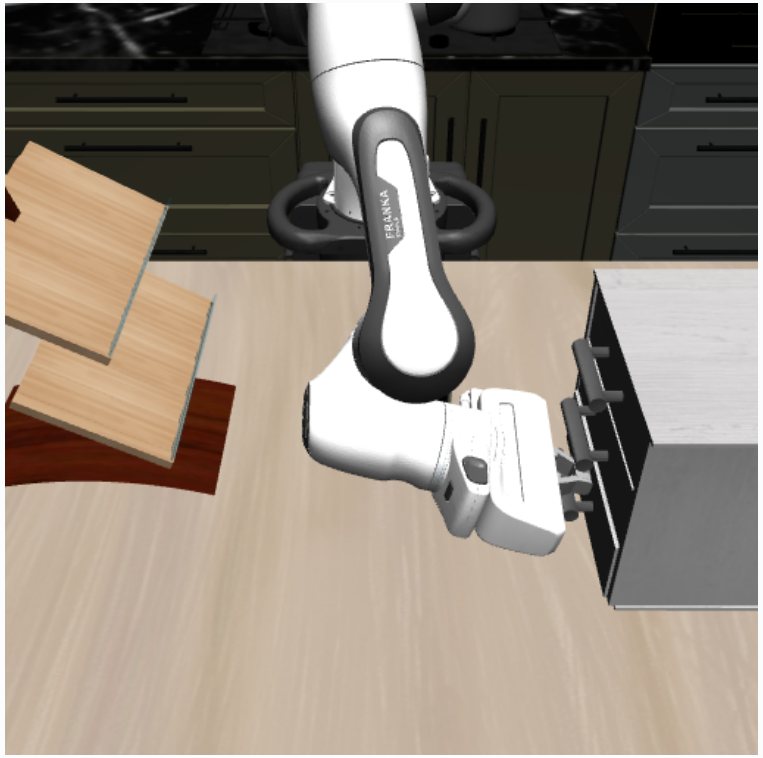
The agent changes its numerical route and hand configuration while fragments execute the repeated control.



Step 0 · Initial scene.
The bowl starts outside the drawer.



Step 393 · Bowl in the cavity.
Model reports a likely slip during transfer.



Step 1202 · Near-final push.
Raw success follows at step 1207.

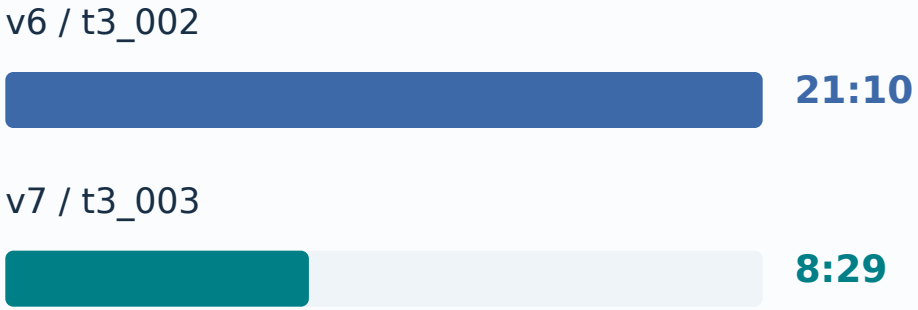
[R2] Unaltered saved external-camera observations. The transfer should not be described as a verified controlled placement.

MEASURED RESULT

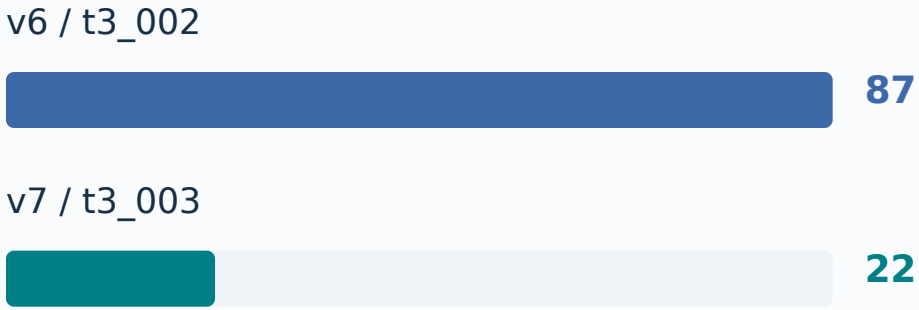
t3: fewer calls, faster raw-goal completion

Net control time falls 59.9%, while the physical action count grows 2.88×.

Video-excluded control time



Model decisions



Official goal succeeds in both

Actions: 419 → 1,207.
Single-run observed time ratio: 2.49×.
No multi-seed reliability claim.

Manipulation quality is unresolved

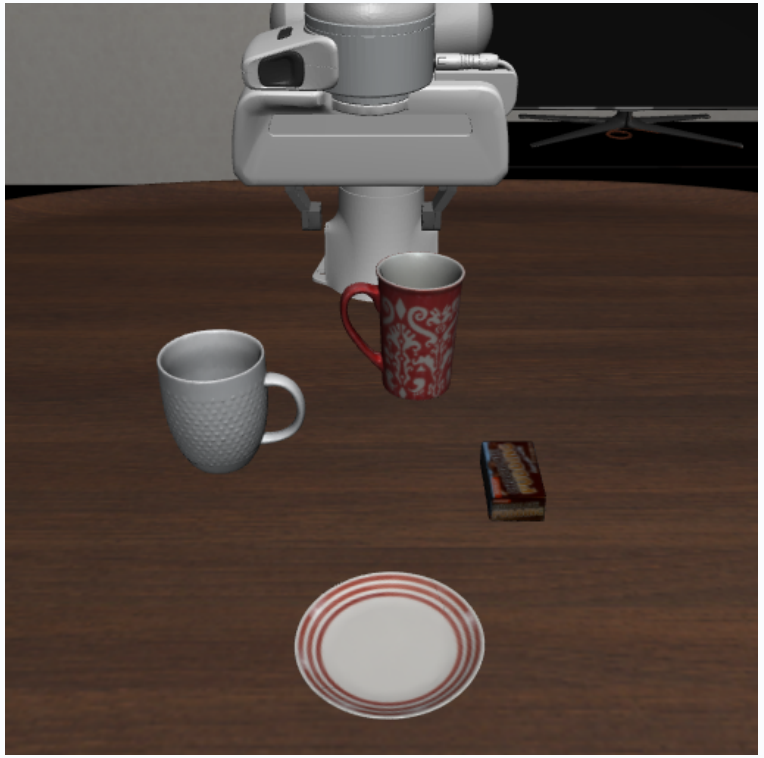
Strict final-state success: false in both.
Contact points: 671 → 1,488.
Final motion/contact has not settled.

[R1,R2] Contact points are accumulated physics samples, not distinct collisions. Counterfactual settle checks are not agent actions.

RECORDED SCENE / T6_009

t6: mug placement and pudding recovery

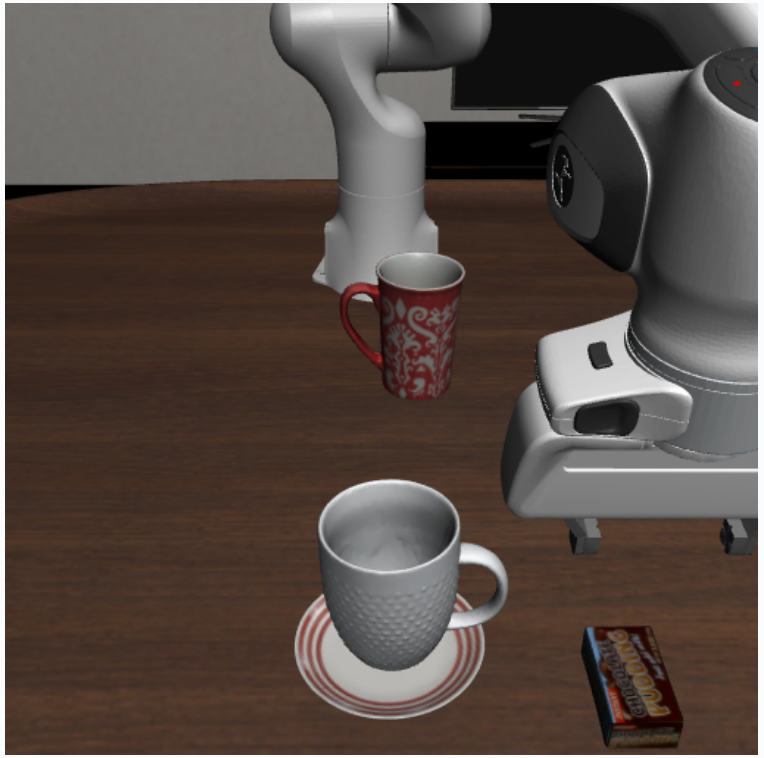
The mug phase finishes in six model decisions; subsequent work is spent on pudding grasp and placement.



Step 0 · Initial scene.
White mug, plate and pudding.



Step 376 · Mug released on plate.
Gripper has withdrawn upward.



Step 1543 · Before user stop.
Pudding released; target relation still false.

[R4] Saved external-camera observations. Final physical audit confirms the mug goal; pudding region membership fails.

EXECUTION EVIDENCE

t6: useful recovery, incomplete task

Stopped by the user at 1,544 steps and 28 model decisions; net elapsed control time 12:35.

1 **Mug transfer**

Approach, close, test-lift, carry, lower, release and withdraw in six decisions.

2 **Pudding descent blocked**

Fingers contact the tabletop. The fragment yields on stall; the model initially suspects a joint limit.

3 **Grasp recovered**

After another blocked descent, close at the reached height and test-lift. The pudding follows.

4 **Placement revised**

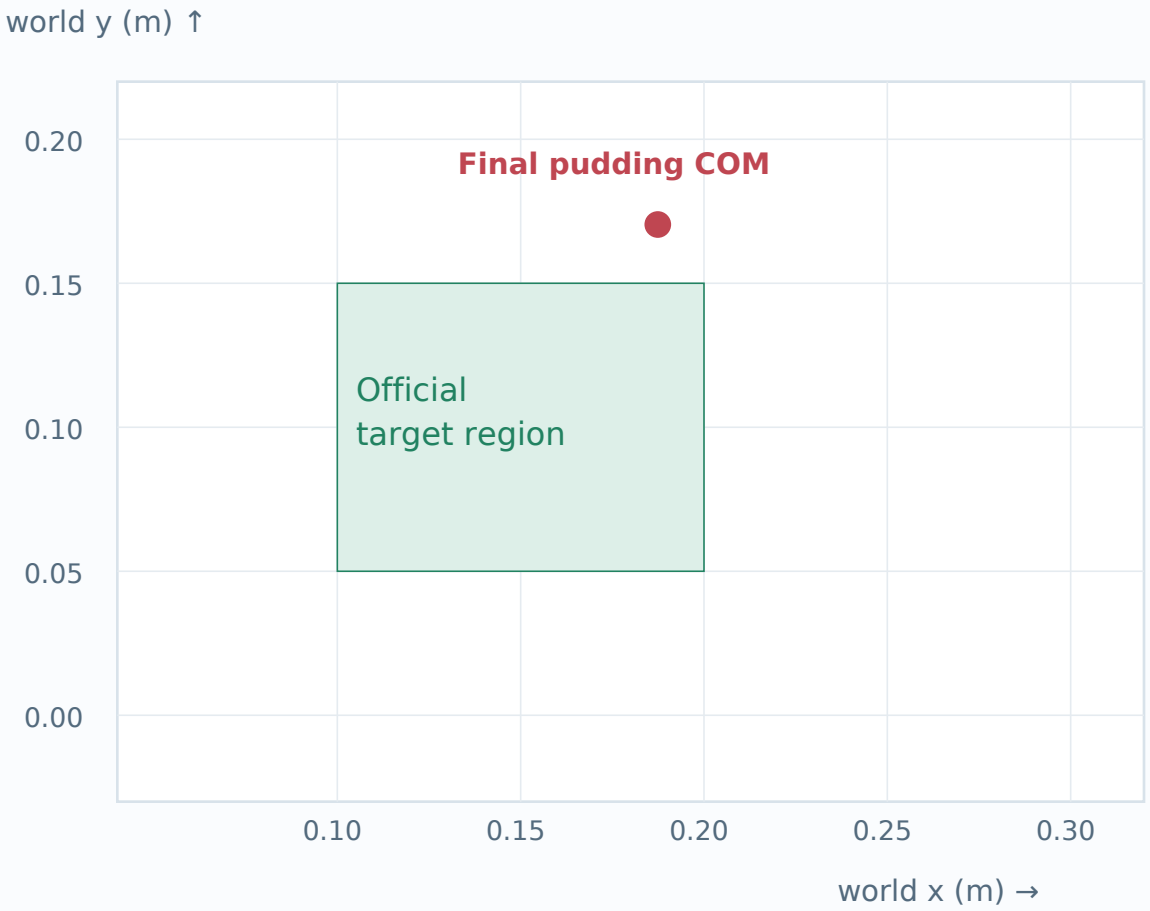
Regrasp and reposition; change wrist pitch to improve reach. Final center still misses the goal region.

[R4] 1,208 forbidden contact-point samples, from fingers/table. Prior t6_007 completed raw goal in 21:29 / 84 calls; no speed ratio here.

GENERAL FRAMEWORK LIMITATION

t6 exposes a coordinate-frame mismatch

Framework right_of = world +x is not the same as the benchmark’s named plate-right region.



Semantic issue

Shared instructions tie “right” to +x. Initial placements follow that assumption, and later recovery must work around it.

Terminal result

Pudding COM ≈ (0.1874, 0.1704) m.
x is within the official range.
y is about **20.4 mm beyond** its upper edge.

Directions need an explicit, task-grounded reference frame.

[R4; M6; BDDL] Evaluator-only post-run diagram, not policy input. Region x=[0.10,0.20], y=[0.05,0.15] m.

CONTROL ATTRIBUTION

Where the framework actually did the work

The model still selects targets. The fragment controller supplies almost all per-step motor commands.

t3_003 — **99.17%**



1,197 fragment actions / **10** direct actions / **0** native macro actions
20 execute grants; 1,207 total control steps.

t6_009 — **99.61%**



1,538 fragment actions / **6** direct actions / **0** native macro actions
26 execute grants; 1,544 total control steps.

The useful combination: model-level perception, planning and recovery + reusable measured execution. These percentages measure command attribution, not the fraction of intelligence supplied by either side.

[R2,R4 agent_stats] Generic v7 fragments are distinct from old semantic macros. Direct counts can include observation/finish holds.

DESIGN CONCLUSIONS

What to retain, and what to improve next

These are conclusions from the recorded tests, not new changes or validated future performance.

Retain

Model ownership of the full task.

Longer measured fragments where targets are clear.

Explicit gripper stages, checkpoints and cause-preserving feedback.

Separate task outcome, physical quality and execution timing.

Next general improvements

Ground directional relations in an explicit frame instead of a universal axis convention.

Verify whole-object placement from new observations.

Improve contact/cause feedback and route quality.

Evaluate multiple tasks and resets before claiming generality.

The English UI is a language-only update. The recorded v7 controller and evaluator were not changed for these slides.

[M1–M6; R1–R5] t3 is a raw-goal timing gain with quality caveats. t6 remains a stopped, incomplete attempt.

APPENDIX

Evidence, timing and reproducibility

Saved recordings only. No new simulator episode, model request or smoke test was used to build this deck.

Run	Outcome	Control / calls	Draft call / review
t3_002 · v6	Raw success; strict false	21:10 / 87 decisions	1 request; manual review
t3_003 · v7	Raw success; strict false	8:29 / 22 decisions	21.4 s; auto-approved
t6_007 · v6	Raw success; strict false	21:29 / 84 decisions	1 request; manual review
t6_009 · v7	User stop; raw/strict false	12:35 / 28 decisions	24.3 s; auto-approved

Preparation interruption: t6_008 was stopped by the user while the first draft request was still pending, before policy actions. It is preserved separately and excluded from control-time ratios.

Source index, exact metrics, reset hashes and frame provenance: **evidence.json**.
Method files: agent.py, agent_client.py, fragments.py, agent_macros.py, spatial.py and the user-check runner.

Full preparation: t3_003 27.1 s; t6_009 32.3 s. Draft-call latency is a subset. Exact timing and source paths: evidence.json.

Source map

All claims are grounded in current source or preserved task artifacts; frame provenance is recorded in evidence.json.

Method source IDs

Directory: src/gpt_field/
Common prefix: agent_speedup_v2_user_check

M1 _agent.py — executive and shared state

M2 _agent_client.py — model interface

M3 _fragments.py — measured execution

M4 _sim.py — observations and evaluation bridge

M5 _agent_macros.py — optional native macros

M6 _spatial.py — directional semantics

UI .html — English plan review interface

Recorded run IDs

Directory: runs/
Prefix: agent_speedup_v2_user_check_libero_10_

R1 t3_002 — prior v6 reference

R2 t3_003 — current v7 task result

R3 t6_007 — prior v6 reference

R4 t6_009 — current v7 user-stopped attempt

R5 t6_008_review — interrupted preparation

Artifacts: summary.json, agent_stats.json, physical_truth.json, review.json and requests/.

BDDL: LIBERO-10 LIVING_ROOM_SCENE6 white-mug / pudding task, plate_right_region (lines 16–24). See evidence.json for its full path.